

Chapter 3

Advance: Perplexity's Relation to Entropy

Group 2

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- **Recalling the Formula:**

$$PP(W) = P(w_1 w_2 \dots w_N)^{-\frac{1}{N}} = \sqrt[N]{\frac{1}{P(w_1 w_2 \dots w_N)}}$$

- **Beyond a Simple Metric:**

- Perplexity is not just a random heuristic for evaluation.
- It serves as a **normalized version** of the probability assigned to the test set.
- Perplexity arises directly from the information-theoretic concept of cross-entropy.

- **Answering the "Mysteries":**

- Explain the relationship between model performance and entropy.
- Justify the use of inverse probability in the calculation.

Entropy: A Measure of Information

- **Definition:**

- Entropy $H(X)$ measures the amount of information in a random variable X .
- X ranges over the set of outcomes χ (words, letters, etc.) with probability function $p(x)$.

- **Mathematical Formula:**

$$H(X) = - \sum_{x \in \chi} p(x) \log_2 p(x) \quad (3.32)$$

- **Units of Measurement:**

- While any log base can be used, we typically use **log base 2**.
- The resulting value is measured in **bits**.

The Intuition

Entropy represents a **lower bound** on the number of bits required to encode information in an optimal coding scheme.

The Horse Race Example

- **Scenario:**

- You want to bet on one of **8 horses**.
- You need to send a short message to a bookie to identify the horse.

- **Naive Encoding Strategy:**

- Use a standard binary representation for each horse.
- Horse 1: 001; Horse 2: 010; ...; Horse 8: 000
- Each horse is coded with **3 bits**. On average, we send **3 bits per race**.

Question

If the horses do not have an equal chance of winning, can we optimize the message length?

The Horse Race Example

- Given a Non-Uniform Distribution:

Horse 1	1/2	Horse 5	1/64
Horse 2	1/4	Horse 6	1/64
Horse 3	1/8	Horse 7	1/64
Horse 4	1/16	Horse 8	1/64

- Entropy Calculation:

$$H(X) = - \sum_{i=1}^8 p(i) \log_2 p(i) = 2 \text{ bits} \quad (3.33)$$

- Encoding Strategy

- More probable $\rightarrow$ Shorter code: Horse 1 = 0 (1 bit).
- Less probable $\rightarrow$ Longer code: Horse 8 = 111111 (6 bits).

From Single Variables to Sequences

- **Language as a Sequence:**

- Words in a language L form sequences: $W = \{w_1, w_2, \dots, w_n\}$.

- **Entropy of a Sequence:**

- Entropy of a random variable ranging over all sequences of length n :

$$H(w_1, w_2, \dots, w_n) = - \sum_{w_{1:n} \in L} p(w_{1:n}) \log p(w_{1:n}) \quad (3.35)$$

- **Entropy Rate:**

- Also known as **per-word entropy**.
- Measures the average information (bits) carried by each word:

$$\frac{1}{n} H(w_{1:n}) = - \frac{1}{n} \sum_{w_{1:n} \in L} p(w_{1:n}) \log p(w_{1:n}) \quad (3.36)$$

The True Entropy of A Language

To measure the true entropy of a language, we need to consider sequences of infinite length:

$$H(L) = \lim_{n \rightarrow \infty} \frac{1}{n} H(w_{1:n}) = - \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{W \in L} p(w_{1:n}) \log p(w_{1:n}) \quad (3.37)$$

Shannon-McMillan-Breiman Theorem

- **Computational Challenge:**

- Summing over all possible sequences $w_{1:n} \in L$ is mathematically intractable.

- **Shannon-McMillan-Breiman Theorem:**

- If a language is both **stationary** and **ergodic**, we can estimate $H(L)$ using a single, sufficiently long sequence:

$$H(L) = \lim_{n \rightarrow \infty} -\frac{1}{n} \log p(w_1, w_2, \dots, w_n) \quad (3.38)$$

- **Concept of Stationarity:**

- Probabilities are invariant to shifts in time.
- N-grams are stationary: P_i depends only on P_{i-1} (for bigrams).

The Reality of Natural Language

- Natural language is **not stationary**.
- Statistical models (N-grams) assume stationarity as a convenient approximation for tractable computation.

- **The Reality of Modeling:**

- We don't know the actual probability distribution p of a language.
- We use a model m as an approximation of p .

- **Formal Definition:**

$$H(p, m) = \lim_{n \rightarrow \infty} -\frac{1}{n} \sum_{w \in L} p(w_1, \dots, w_n) \log m(w_1, \dots, w_n) \quad (3.39)$$

- Applying the Shannon-McMillan-Breiman theorem, for a stationary ergodic process:

$$H(p, m) = \lim_{n \rightarrow \infty} -\frac{1}{n} \log m(w_1, w_2, \dots, w_n) \quad (3.40)$$

- **The Fundamental Inequality:**

- For any model m , the cross-entropy $H(p, m)$ is an upper bound on the true entropy $H(p)$:

$$H(p) \leq H(p, m) \quad (3.41)$$

- **Measuring Model Accuracy:**

- The gap between $H(p, m)$ and $H(p)$ represents the inaccuracy of our model.
- The more accurate m is, the closer $H(p, m)$ gets to $H(p)$.

Perplexity's Relation to Entropy

- **Approximating the Limit:**

- While cross-entropy is defined in the limit ($n \rightarrow \infty$), we approximate it using a sufficiently long sequence of fixed length N :

$$H(W) = -\frac{1}{N} \log P(w_1 w_2 \dots w_N) \quad (3.42)$$

- **Formal Definition of Perplexity:**

- Perplexity is defined as 2 raised to the power of the cross-entropy:

$$PP(W) = 2^{H(W)}$$

- Expanding $H(W)$ back into the perplexity formula:

$$PP(W) = P(w_1 w_2 \dots w_N)^{-\frac{1}{N}} = \sqrt[N]{\frac{1}{P(w_1 w_2 \dots w_N)}}$$

- What is the fundamental relationship between the cross-entropy $H(p, m)$ of a model and the true entropy $H(p)$ of the actual language?
- Why must we take the limit as $n \rightarrow \infty$ in equation (3.37) to calculate the true entropy of a language $H(L)$?